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S. No. of Question Paper : 6508

Unique Paper Code : 32341502

HC

Name of the Paper : Theory of Computation

Name of the Course : B.Sc. (II) Computer Science

Semester : V.

Duration : 3 Hours

Maximum Marks : 75

(Write your Roll No. on the top immediately on receipt of this question paper.)

All questions from Part A are compulsory.

Attempt any four questions from Part B.

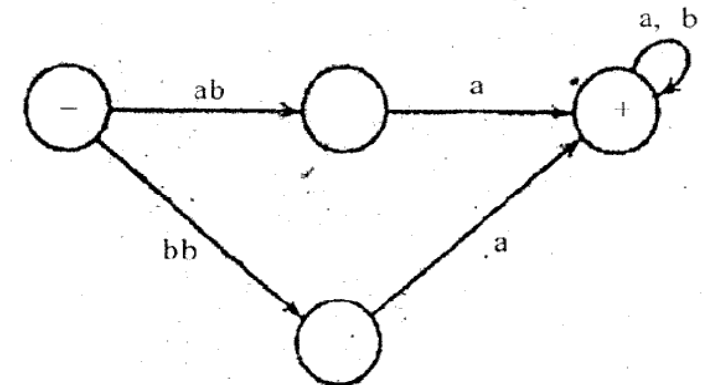
Assume $\Sigma = \{a, b\}$ is the underlying alphabet unless mentioned otherwise. Parts of a question must be answered together.

Part A

1. (a) Consider the language S^* , where $S = \{aa, b\}$. How many words does this language have of length 4 ? of length 5 ? of length 6 ? What can be said in general ? 2

P.T.O.

- (b) Let $S = \{ab, bb\}$ and let $T = \{ab, bb, bbbb\}$. Show that $S^* = T^*$. 2
- (c) Give a regular expression for the language of all the words that do not have 'aa' as substring. 3
- (d) Generate a CFG for b^*a^* . 3
- (e) Design a Deterministic Finite Automata for the language of all the words that end in a double letter. 4
- (f) Using Pumping Lemma, prove that language $a^n b^{2n}$, $n \geq 0$ is not regular. 4
- (g) Convert the following Transition Graph into its equivalent Regular Expression : 4



(h) Show that the complement of a recursive language is also recursive. 4

(i) Construct a Push Down Automata for $a^n b^{n+1}$ where $n \geq 1$. 4

(j) If $L_1 = (a+b)b(a+b)^*$ and $L_2 = (a+b)^*b$, find a Regular Expression and Deterministic Finite Automata for $L_1 \cap L_2$. 5

Part B

2. (a) Begin with the grammar : 5

$S \rightarrow ABC|BaB$

$A \rightarrow aA|BaC|aaa$

$B \rightarrow bBb|a|D$

$C \rightarrow CA|AC$

$D \rightarrow \epsilon$.

P.T.O.

(i) Eliminate ϵ productions.

(ii) Eliminate any unit productions in the resulting grammar.

(iii) Eliminate any useless symbols in the resulting grammar.

(b) Using Pumping Lemma, prove that language $a^n b^n a^n$, $n \geq 1$ is non-context free. 5

3. (a) Prove that the regular languages are closed under complement. 3

(b) Give a CFG for the language of all the words having 'bbb' as substring. 3

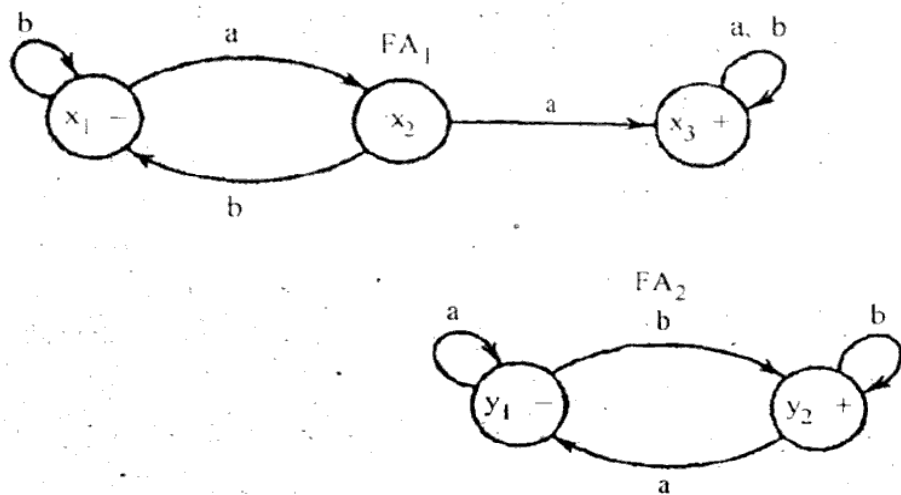
(c) Show that the following grammar is ambiguous : 4

$S \rightarrow XbaaX|aX$

$X \rightarrow Xa|Xb|\wedge$.

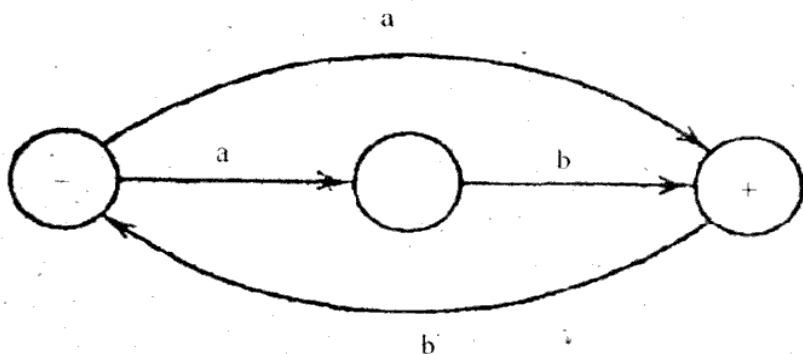
4. (a) Find $FA_1 FA_2$ (concatenation) for the following automata :

5



- (b) Find the equivalent Deterministic Finite Automata for the following Non-deterministic Finite Automata :

5



P.T.O.

5. (a) Describe the language for the following regular expressions :

4

(i) $(a+b)^* ab(a+b)^*$

(ii) $((a+b)b)^*$

- (b) Build a DFA that accepts all words with fewer than four letters.

3

- (c) Give a regular expression for the language of all the words that do not have a double letter.

3

6. (a) Explain halting problem.

2

- (b) Show that if language L is recursive, then L is recursively enumerable also.

3

- (c) Design a Turing Machine for $a^n b^n$ for $n \geq 1$.

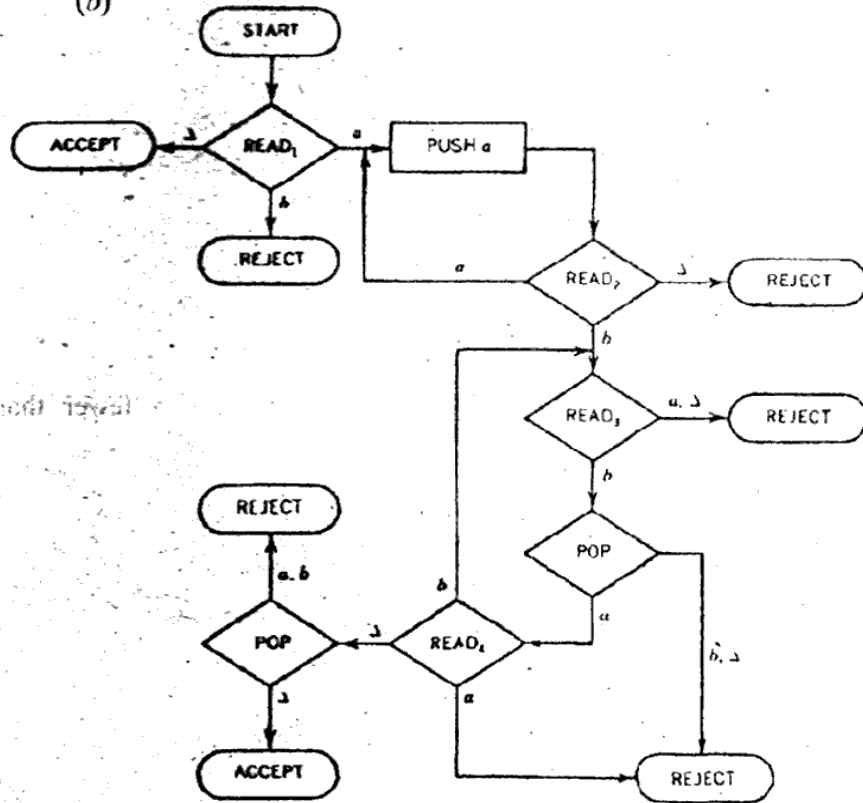
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7. (a) Design a PDA for the following language

5

$$L = \{a^n S, \text{ where } S \text{ starts with } b \text{ and } \text{length}(S) = n\}$$

(b)



(i) Define the language defined by above PDA. 2

(ii) Trace the word 'abb' on the above PDA. 3